Project Proposal

VICOO – Sustainable Fashion Platform

COMP3030J

Group 7: SixSeven

Date: April 3, 2026

Who We Are — VICOO

VICOO is a six-member software engineering team specialising in **full-stack web platform development** that integrates commercial viability with **measurable social impact**. Our technical expertise spans distributed systems, database engineering, networking, and modern frontend technologies.

The Client

Our client is a **fast-fashion retail brand** undergoing a strategic transition toward responsible and sustainable business practices. Facing **increasing consumer scrutiny** and **ESG (Environmental, Social, Governance)** regulatory pressure, the client requires a technology solution to **transform sustainability** from a marketing narrative into a verifiable and transparent system.

The Problem

The global fashion industry contributes about **10% of annual carbon emissions** and faces three key structural hurdles: **opaque supply chains, one-directional charity and overlooked community creativity**. Its sustainability efforts align with UN SDGs, prioritizing **SDG 12 (Responsible Consumption & Production)**, with SDG 1,13,17 as secondary focus.

Our Solution & Vision

VICOO launches a **two-module e-commerce platform** combining sustainability transparency and social co-creation. **Module A (Cotton Source Traceability)** offers supplier & batch registration, QR-code product traceability and fewer intermediaries to cut emissions. **Module B (Community Co-Creation)** features a rural children's artwork submission platform, public voting for product design, and fully traceable donation tracking for each purchase.

Stakeholder Benefits

- **Consumers**: Full supply chain transparency and participation in charity.
- **Client Brand**: Strong ESG credibility and market differentiation.
- **Rural Children**: Monetisation opportunities and social recognition.
- **Society**: Reduced waste and scalable sustainability model.
- **Academic Goals**: Full software engineering lifecycle implementation.

Deliverables

- **Platform Core**: Authentication, product management, order system, file storage.
- **Module A**: Supply chain tracking, QR traceability, carbon estimator.
- **Module B**: Artwork submission, voting system, donation tracker.
- **Admin Panel**: User management and analytics dashboard.
- **Technical Documentation**: Architecture, API, database schema, deployment manual.
- **Live System**: Fully deployed platform accessible remotely.

Success Criteria (SMART)

- **Supply Chain Transparency**: ≥80% traceable products by Week 7.
- **Charity Co-Creation Pipeline**: Functional workflow with ≥30% voting participation by Week 8.
- **Platform Stability**: Core system functionalities tested and operational by Week 9.
- **Remote Accessibility**: Fully deployed and accessible system by Week 10.

Technology Stack

- **Frontend**: React + TypeScript + Tailwind CSS for rapid UI development.
- **Backend**: Flask (Python) for lightweight, flexible server-side logic.
- **Database**: MySQL 8.x for structured relational data storage.
- **Deployment**: Docker + Course VM for reproducible, accessible deployment.

Remote Testing Strategy

The system will be containerised using Docker and deployed on the course VM.

Budget (Details contained in the appendix)

Total Estimated Budget: $12,000.

Why Choose VICOO?

VICOO's solution doesn't just add a sustainability label but redefines **fast-fashion operations** at its core: all products are **traceable**, all donations **auditable**, and all community **contributions valued**. We will deliver a fully functional system in a realistic 12-week timeline, built on software engineering best practices and **aligned with global sustainability goals**.

Appendix：

Budget Breakdown：

Expense Category	Estimated Cost	Details
Platform Development Costs	**$5,000**	Front-end and back-end development, database setup, and course VM hosting fees.
API Integration Costs	**$3,000**	**Cotton Traceability API**: $1,500 annually **QR Code & Donation Tracking API**: $1,500 annually
Content Creation & Editing	**$1,500**	Writing, editing, and researching sustainability and rural children's artwork content.
Marketing & Promotion	**$1,000**	Social media promotion, ESG branding materials, and community outreach.
Platform Maintenance & Updates	**$1,000**	Server hosting, bug fixes, feature updates, and technical support.
Miscellaneous Expenses	**$500**	Project management tools, graphic design software, and unforeseen costs.
Total Estimated Budget	**$12,000**	

Team qualifications:

Member	Role	Core Responsibilities
Huang Tian	Backend Lead & Technical Coordinator	Lead backend MVC architecture design and development; formulate API interface specifications.
Yang Haoze	Product Designer & System Architect	Product prototyping; UI/UX visual and interaction specifications; functional process design.
Li Jinhao	Database Lead	Database table structure design; ER modeling; data storage and query optimization.
Wang Jundi	Backend Developer & System Integration	Backend functional module development; third-party API integration.
Liu Honghao	Frontend Developer	Full page development for client and admin ends; implementation of interactive effects.
Ma Qingchuan	Test Engineer & Documentation Lead	Full-system functional testing; test reports and user operation manuals.

Project schedule & deliverables:

Group Name	Team Members	March				April				May				June				
		W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	Notes
WP 1: System Design & UI/UX Manager: Yang Haoze	Liu Honghao																	
Task 1.1: Design structure	Huang Tian																	
Task 1.2: Core features																		
Task 1.3: UI/UX Design																		
Task 1.4: Support development																		
WP 2: Database Manager: Li Jinhao	Wang Jundi																	
Task 2.1: Schema Design	Ma Qingchuan																	
Task 2.2: Closed-loop Logic	Huang Tian																	
Task 2.3: Integration																		
Task 2.4: Performance																		
WP 3: Frontend Manager: Liu Honghao	Yang Haoze																	
Task 3.1: UI Structure & Nav																		
Task 3.2: Core Page Dev																		
Task 3.3: Map & Viz																		
Task 3.4: QA & Polish																		
WP 4: Backend Manager: Huang Tian	Ma Qingchuan																	
Task 4.1: Core Backend Logic	Wang Jundi																	
Task 4.2: DB Integration	Li Jinhao																	
Task 4.3: Map API Processing	Yang Haoze																	
Task 4.4: API Refinement																		
WP 5: Integration Manager: Wang Jundi	Yang Haoze																	
Task 5.1: Standards & Protocol	Huang Tian																	
Task 5.2: Joint Debugging	Liu Honghao																	
Task 5.3: Optimization	Li Jinhao																	
Task 5.4: Validation	Ma Qingchuan																	
Task 5.5: Final Review																		
WP 6: Testing & QA Manager: Ma Qingchuan	Wang Jundi																	
Task 6.1: Test Planning	Li Jinhao																	
Task 6.2: System Testing																		
Task 6.3: Optimization																		
Task 6.4: Documentation																		

Deliverable
Milestone

Work Package:

PROJECT / GROUP NAME	**Group 7**		
Start Date	2026/03/16 (Week 3)	**Finish Date**	2026/05/18 (Week 16)
Aim / Objective	To design the overall system architecture, define the core features and user flows, and create UI/UX designs that support the successful development and integration of the application.		
Work package Manager	Yang Haoze		
Contributors to this package	Liu Honghao Huang Tian		
Description / Activities	Task 1.1 Design the overall structure of the system and define the relationship between major components. • 1.1.1 Identify the main components of the system • 1.1.2 Define how the frontend, backend, and database interact Task 1.2 Define the core features of the application and organise them based on project requirements. Task 1.3 Design the user interface and improve the user experience of the application. • 1.3.1 Create wireframes for key pages • 1.3.2 Refine navigation and user flow Task X.4 Support the development process by reviewing implementation and ensuring consistency with the original design.		
Milestones		Week	
	M 1.1 Completion of the initial system structure and project feature framework.	Week 3	
	M 1.2 Completion of core user flow design and functional module planning.	Week 5	
	M 1.3 Architecture design, key interface design, and feature plan ready for the interim report.	Week 8	
	M 1.4 Refinement of system design based on development progress and team feedback.	Week 11	
	M 1.5 Completion of the final review of system structure, feature consistency, and interface logic.	Week 13	
	M 1.6 Final release support completed, with the design and architecture fully reflected in the implemented system.	Week 16	
Deliverables		Week	
	D 1.1 Architecture and module design document.	Week 3	
	D 1.2 Core feature and user journey specification.	Week 5	
	D 1.3 Key interface wireframes and interaction design.	Week 8	
	D 1.4 Revised design document based on implementation feedback.	Week 11	
	D 1.5 Final system and interface review document.	Week 13	
	D 1.6 Final design and implementation support document.	Week 16	

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<tr><td>PROJECT / GROUP NAME</td><td colspan="3">Group 7</td></tr>
<tr><td>Start Date</td><td>(Week 4)</td><td>Finish Date</td><td>Week 16</td></tr>
<tr><td>Aim / Objective</td><td colspan="3">1. To design and maintain a robust database schema that supports the "Closed-loop Charity" model.
2. To ensure high data integrity for children's art voting and cotton sourcing information.
3. To provide seamless data modeling for the backend calculation engine and frontend gallery.</td></tr>
<tr><td>Work package Manager</td><td colspan="3">Li Jinhao</td></tr>
<tr><td>Contributors to this package</td><td colspan="3">Wang Jundi, Ma Qingchuan, Huang Tian</td></tr>
<tr><td>Description / Activities</td><td colspan="3">Task 2.1 Database Schema Design & ER Modeling Architect the foundational data structure by defining entities for user profiles, children's artworks, voting logs, and supply chain coordinates to ensure data consistency across the platform.

2.1.1 Conduct a comparative analysis of data requirements for stakeholders (charities, donors, and suppliers).

2.1.2 Create high-fidelity Entity-Relationship (ER) diagrams to visualize data flows between the voting system and donation engine.

Task 2.2 Finalizing "Closed-loop Charity" Data Logic Design the data models for the core "Fast Fashion Charity" logic, specifically structuring the tables required for tracking the transparent 30% donation split and premium cotton sourcing.

2.2.1 Define the logic for storing and retrieving raw material sourcing coordinates (premium cotton origins) for Map API integration.

2.2.2 Implement the database constraints for the donation tracking system to ensure financial transparency.

Task 2.3 Database-Backend Integration & Data Binding Collaborate with the Backend Lead to align database models with server-side logic, ensuring seamless data retrieval for the Children's Gallery and Voting interface.

2.3.1 Develop structured data endpoints in coordination with the API refinement process for optimized JSON responses.

2.3.2 Store and retrieve children's art voting records while maintaining data integrity for the frontend display.

Task 2.4 Performance Optimization & QA Support Conduct query optimization and support the Testing Lead during the end-to-end validation of the "Transparency Loop" to identify and resolve data-related bottlenecks.

2.4.1 Validate data integrity during integration testing between frontend</td></tr>
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	and backend components. **2.4.2** Assist in compiling the technical documentation, specifically focusing on the database schema and maintenance manual.	
Milestones		Week
	M 2.1 Finalization of DatabaseSchema	Week 5
	M 2.2 Data binding forvoting/gallery completed	Week 6
	M 2.3 Full integration of donation tracking data	Week 9
	M 2.4 Database performance tuning completed	Week 13
	M 2.5 Final QA Report contribution completed	Week 16
Deliverables		Week
	D 2.1 DB Schema &ER Diagram	Week 5
	D 2.2 Data Integration Report	Week 6
	D 2.3 Core Functional Dataset	Week 9
	D 2.4 Bug Tracking & Data Fix Report	Week 13
	D 2.5 Final Project Package (DB Manual)	Week 16

PROJECT / GROUP NAME	Group 7 - Fast Fashion Charity (Frontend Developer)		
Start Date	2026/03/23 (Week 4)	**Finish Date**	2026/04/26 (Week8)
Aim / Objective	To design and implement the front-end of the platform with a clean, user-friendly, and responsive interface, enabling users to browse the Children's Gallery, explore the Supply Chain Map (cotton origin visualization), participate in voting, and understand the transparency loop (design → vote → donation tracking) through clear UI components.		
Work package Manager	Liu Honghao (Frontend Developer)		
Contributors to this package	Yang Haoze		
Description / Activities	Task 3.1 UI Structure & Navigation (Information Architecture) Design the overall layout and page flow for the platform. • 3.1.1 Define the main navigation (Home / Gallery / Map / Vote / Donation Transparency). • 3.1.2 Create consistent page templates (headers, footers, card grids, detail pages). • 3.1.3 Establish a unified visual style (color palette, typography, spacing, accessibility). Task 3.2 Core Page Development (Front-End Implementation) Build the core pages with reusable components. • 3.2.1 Implement the Children's Gallery page (artwork grid + detail modal/page). • 3.2.2 Implement the Voting Interface (vote actions, vote states, feedback prompts). • 3.2.3 Implement the Donation Transparency page (progress indicators, breakdown panels). Task 3.3 Supply Chain Map & Data Visualization Deliver the interactive map experience and cotton origin display. • 3.3.1 Implement a map container and origin markers (Map API integration-ready UI). • 3.3.2 Design map overlays (origin details, sourcing notes, traceability tags). • 3.3.3 Ensure map interactions are mobile-friendly (zoom, pan, marker selection). Task 3.4 Front-End Quality Assurance & UX Polish Validate usability, consistency, and performance across devices. • 3.4.1 Responsive testing for key breakpoints (mobile/tablet/desktop). • 3.4.2 Fix UI bugs and edge cases (empty states, loading states, error states). • 3.4.3 Improve perceived performance (lazy loading images, skeleton UI, caching hints).		
Milestones			Week
	M 3.1 Main UI structure & navigation completed		Week 4
	M 3.2 Core pages (Gallery + Vote + Transparency) completed		Week 6
	M 3.3 Supply Chain Map UI completed		Week 7
	M 3.4 Front-end testing & UX refinement completed | Week 8		Week 8

Deliverables		Week
	D 3.1 UI/UX prototype or wireframe set (page flow + component spec)	Week 4
	D 3.2 Implemented front-end core pages (Gallery / Vote / Transparency)	Week 6
	D 3.3 Supply Chain Map interface (interactive map + origin detail panels)	Week 7
	D 3.4 Final front-end build (responsive + bug-fixed + demo-ready)	Week 8

PROJECT / GROUP NAME	Group 7 - Fast Fashion Charity (Backend & Logic Integration)	
Start Date	2026/03/18 (Week 4)	**Finish Date** 2026/04/28 (Week 9)
Aim / Objective	To lead the development of the 'Fast Fashion Charity' core backend logic, focusing on the transparent 30% donation calculation engine and Map API data processing. Collaborate closely with the Database and Frontend leads to ensure seamless system-wide data flow and integration.	
Work package Manager	Huang Tian (Backend Lead & Integration Collaborator)	
Contributors to this package	Wang Jundi, Ma Qingchuan, Li Jinhao	
Description / Activities	Task 4.1 Core Backend Logic & Charity Engine (Weeks 4-5) Architect and implement the 'Closed-loop Charity' calculation engine. This involves developing the server-side logic that computes the 30% donation split from apparel sales and tracks it through the transparency loop. Task 4.2 Database Integration & Data Utilization (Weeks 5-6) Collaborate with the Database Lead to align backend models with the schema. Utilize database information to store and retrieve children's art voting records and cotton sourcing data, ensuring data integrity for the frontend gallery. Task 4.3 Map API Processing & Sourcing Visualization Logic (Weeks 6-7) Develop backend handlers for Map API integration. Process raw material sourcing coordinates (premium cotton origins) and expose them as structured API endpoints to support the frontend 'Supply Chain Map' visualization. Task 4.4 API Refinement & System-Wide Integration (Weeks 8-9) Conduct collaborative API refinement with the Frontend Lead to optimize JSON responses for the voting system. Support the Testing Lead during the 'Transparency Loop' end-to-end validation, focusing on backend performance and error handling.	
Milestones		Week
	M 4.1 Backend environment & Core Charity logic engine completed	Week 5
	M 4.2 Successful database-backend data binding for voting records	Week 6
	M 4.3 Map-based sourcing data API fully functional for frontend use	Week 7
	M 4.4 Comprehensive API suite finalized for the voting & donation modules	Week 8
	M 4.5 Successful collaborative integration & backend performance tuning	Week 9
Deliverables		Week
	D 4.1 Backend Architecture Spec (including Charity Calculation Logic)	Week 5
	D 4.2 Data Integration Report (Backend-to-Database mapping)	Week 6
	D 4.3 Functional API Codebase (Sourcing Map & Voting Endpoints)	Week 8
	D 4.4 Technical Documentation (API Specs & Integration Guide)	Week 8
	D 4.5 Backend Refinement Report (Post-Integration & Test Support)	Week 9

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<tr><td>PROJECT / GROUP NAME</td><td colspan="3">Group 7</td></tr>
<tr><td>Start Date</td><td>(Week 4)</td><td>Finish Date</td><td>Week 16</td></tr>
<tr><td>Aim / Objective</td><td colspan="3">1. To lead the end-to-end Frontend-Backend integration for the Fast Fashion Charity project, establishing standardized and seamless data interaction protocols between frontend and backend systems.
2. To ensure the consistency and reliability of API communication, optimize interface response efficiency, and resolve all cross-system integration issues.
3. To collaborate with cross-functional leads to align integration standards, support end-to-end validation of the Transparency Loop, and guarantee the stable operation of the whole system.</td></tr>
<tr><td>Work package Manager</td><td colspan="3">Wang Jundi</td></tr>
<tr><td>Contributors to this package</td><td colspan="3">Huang Tian, Yang Haoze, Liu Honghao, Li Jinhao, Ma Qingchuan</td></tr>
<tr><td>Description / Activities</td><td colspan="3">Task 5.1 Integration Protocol & API Standard Definition (Weeks 4-5)
Establish unified Frontend-Backend integration protocols and API design standards for the project, including JSON response formatting, request parameter rules, and error code systems.

5.1.1 Collaborate with the Backend Lead to sort out all backend API endpoints and define interface calling specifications for frontend development.

5.1.2 Create an integration interface document to standardize the interaction process between frontend and backend modules (voting system, donation engine, supply chain map).

Task 5.2 Frontend-Backend Interface Alignment & Joint Debugging (Weeks 5-7)
Lead the joint debugging of core business modules, align frontend data requests with backend API responses, and resolve interface mismatches, data transmission errors and cross-domain issues.

5.2.1 Coordinate with the Backend Lead to debug the Charity calculation engine and voting system APIs, ensuring accurate data transfer for donation split and voting records.

5.2.2 Collaborate with the Database Lead and Backend Lead to verify the consistency of sourced data (premium cotton origins) for the Map API, supporting frontend supply chain visualization.

Task 5.3 Integration Optimization & Performance Tuning (Weeks 7-8)
Optimize the Frontend-Backend integration link, reduce API response latency, and optimize data transmission efficiency for high-frequency calling interfaces (e.g., real-time voting, donation tracking).

5.3.1 Refine JSON response structures to remove redundant data and reduce data transmission volume between frontend and backend.

5.3.2 Conduct pressure testing on core integration interfaces and resolve</td></tr>
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	performance bottlenecks in the interaction process. **Task 5.4 End-to-End Integration Validation & Issue Resolution (Weeks 8-9)** Support the project's end-to-end Transparency Loop validation, conduct full-system integration testing, and track and resolve all Frontend-Backend integration-related bugs and issues. 5.4.1 Collaborate with the Testing Lead to execute integration test cases and verify the normal operation of all cross-module business processes. 5.4.2 Compile an integration issue tracking and resolution report, and provide final optimization suggestions for the Frontend-Backend interaction system.	
Milestones		Week
	M 5.1 Frontend-Backend Integration Protocol & API Standard Document finalized	Week 5
	M 5.2 Core module interface joint debugging completed, no critical integration errors	Week 6
	M 5.3 Integration link optimized, all core APIs meet performance requirements	Week 9
	M 5.4 Full-system integration test cases executed, major integration issues resolved	Week 13
	M 5.5 End-to-end Transparency Loop integration validated, whole system runs stably	Week 16
Deliverables		Week
	D 5.1 Frontend-Backend Integration Protocol & API Standard Specification	Week 5
	D 5.2 Interface Joint Debugging Report & Core Module Integration Log	Week 6
	D 5.3 Integration Performance Optimization Report & API Tuning Record	Week 9
	D 5.4 Full-System Integration Test Report & Issue Tracking Sheet	Week 13
	D 5.5 Final Frontend-Backend Integration Summary Document & System Operation Guide	Week 16

PROJECT / GROUP NAME	**Group 7**		
Start Date	Week 9	**Finish Date**	Week 16
Aim / Objective	1.Ensure the reliability, stability, and usability of the system through systematic testing and quality assurance processes. 2.Identify and resolve bugs and performance issues in the website before the final release. 3.Produce clear testing documentation and assist in maintaining complete project documentation for the final submission.		
Work package Manager	Ma Qingchuan		
Contributors to this package	Wang Jundi, Li Jinhao		
Description / Activities	Task 6.1 Test Planning and QA Strategy 6.1.1 Define the testing scope and objectives for the project. 6.1.2 Design a test plan including functional testing, usability testing, and integration testing. 6.1.3 Prepare testing scenarios and test cases for key system features such as the children's design voting system and donation transparency module. Task 6.2 System Testing and Bug Identification 6.2.1 Perform functional testing to verify that core system functions operate correctly. 6.2.2 Conduct integration testing between frontend and backend components. 6.2.3 Record identified bugs and coordinate with developers to resolve issues. Task 6.3 Website Optimization 6.3.1 Evaluate website performance and usability. 6.3.2 Identify potential issues related to loading speed, navigation, and user interaction. 6.3.3 Work with developers to improve system performance and user experience. Task 6.4 Documentation and QA Reporting 6.4.1 Document testing procedures and results. 6.4.2 Prepare a QA report summarizing testing results and bug fixes. 6.4.3 Assist in compiling project documentation including the final technical report and user manual.		

Milestones		Week
	M 6.1 Test Plan Completion	Week 9
	M 6.2 Initial System Testing	Week 11
	M 6.3 Bug Tracking & Fixing	Week 13
	M 6.4 Final System Testing	Week 15
	M 6.5 QA Report Completion	Week 16

Deliverables		Week
	D 6.1 Test Plan Document	Week 9
	D 6.2 Test Case Collection	Week 11
	D 6.3 Bug Tracking Report	Week 13
	D 6.4 Final QA Testing Report	Week 15
	D 6.5 Documentation Contribution	Week 16